ASHLIN JAISHAL JAYAKUMAR

London • +44 7587 934672 • ashlinjaishal23@gmail.com • linkedin.com/in/ashlin-jayakumar-8272521b3 • https://github.com/AshJai23

MSc Data Science graduate with a Computer Science Engineering background and practical experience across data analysis, business reporting, machine learning, deep learning, AI engineering and prompt engineering. Skilled in Python, SQL, Power BI, Excel, Scikit-learn and TensorFlow/Keras, with experience building analytical pipelines, dashboards, predictive models, API-based systems and data visualisation outputs. Strong project portfolio covering fraud detection, cyber threat detection, time-series forecasting, healthcare prediction, image classification, environmental analytics, regression modelling and blockchain-based security. Combines technical problem-solving with UK customer-facing and leadership experience, bringing communication, adaptability, stakeholder awareness and disciplined delivery to data, AI and software-focused roles.

WORK EXPERIENCE

Starbucks • UK • 04/2026 - Present

Barista • Part-time

- Deliver customer service in a fast-paced cafe environment while maintaining beverage quality, cleanliness and food safety standards.
- Prepare orders accurately, handle POS/till transactions and support smooth store operations during peak periods.
- Strengthen communication, teamwork, multitasking and time management while working with diverse customers and colleagues.

Superdrug • Morden, UK • 05/2024 - 05/2026

Sales Team Lead • Part-time

- Promoted for consistent performance and leadership, supporting store operations, team coordination and customer satisfaction.
- Reviewed sales trends, stock movement and customer behaviour to support visual merchandising and operational decisions.
- Guided team members during busy periods, strengthening leadership, problem-solving, service quality and task prioritisation.

Sales Advisor

- Delivered high-impact customer service in a fast-paced retail environment while maintaining product and process knowledge.
- Supported stock control, campaign execution, sales reporting and accurate POS, returns and store process execution.
- Handled customer queries professionally while building communication, resilience and customer relationship skills.

Nuffield Health • Fulham, UK • 01/2024 - 06/2024

Front Desk Staff

- Managed client bookings, front-desk enquiries and member support while maintaining professionalism and system accuracy.
- Handled confidential customer information carefully and supported smooth front-of-house workflows in a healthcare setting.
- Improved UK workplace communication, organisation, attention to detail and multitasking in a service-focused environment.

Sainsbury's • Wimbledon, UK • 11/2023 - 01/2024

Customer Experience Assistant

- Supported customers during peak trading periods, handling POS transactions, returns and shop-floor queries efficiently.

- Assisted stock control, store presentation and smooth transaction flow in a high-volume retail environment.
- Built resilience, time management and customer-facing confidence while balancing part-time work with MSc studies.

Vision Group • India • 07/2022 - 08/2023

Business Analyst / Data Analyst

- Built SQL and Python data pipelines processing 1M+ records, improving data reliability by 15% and supporting analytics workflows.
- Automated reporting processes using Python/SQL and workflow mapping, reducing reporting turnaround time by 50%.
- Designed Power BI dashboards and structured reports for KPI monitoring, operational analysis and stakeholder decision-making.
- Gathered business requirements, translated them into functional specifications and documented reporting logic and data flows.
- Maintained reporting pipelines with focus on data accuracy, consistency, traceability and audit-ready outputs.
- Presented analytical findings to technical and non-technical stakeholders and supported testing, validation and release documentation.

EDUCATION

MSc Data Science

University of Roehampton • London • 09/2023 - 01/2025

- Thesis / applied research focus: machine learning, predictive analytics and data-driven decision-making using structured and unstructured datasets.
- Modules: Statistics and Probability for statistical foundations, uncertainty, distributions and data interpretation.
- Modules: Machine Learning, Deep Learning, AI and NLP for model development, neural networks, text analytics and intelligent systems.
- Modules: Data Analytics, Data Visualisation and Functions of Data Science for EDA, insight generation, dashboards and data storytelling.
- Modules: Big Data and Time Series for scalable data processing, forecasting, sequence modelling and trend analysis.
- Projects completed across fraud detection, cyber threat detection, stock forecasting, image classification, healthcare prediction and regression modelling.

B.E. Computer Science and Engineering

St. Xavier's Catholic College of Engineering • India • 09/2018 - 07/2022

- Final-year thesis/project: ATM Fraud Prevention using Fingerprint Sensor and Blockchain, focused on biometric authentication and secure transaction logging.
- Core modules: Python, Data Structures and Algorithms, Java, Object-Oriented Programming, DBMS, Cloud Computing and Robotics.
- Additional modules: Green Computing, Professional Ethics, IEEE standards/concepts, Digital Signal Processing and Engineering Mathematics M1, M2, M3 and M4.
- Built foundations in programming logic, database design, algorithms, software engineering, web development and secure system design.

CERTIFICATIONS

Data Analytics Professional Certificate

Google

Cloud Practitioner

AWS

Artificial Intelligence
Hope AI

Python for Data Science and Machine Learning

PROJECTS

Deep Learning Fraud Detection System
- Developed a CNN-LSTM fraud detection model to identify suspicious transaction patterns from structured data workflows.
- Achieved 99.2% recall, prioritising fraud capture and reducing the risk of missed suspicious cases.
- Reduced false positives by 30% through feature engineering, anomaly detection logic and model tuning.
- Built preprocessing steps for data cleaning, scaling, feature preparation and model-ready dataset generation.
- Used Scikit-learn metrics including recall, precision, F1-score and confusion matrix to evaluate results.
- Prepared monitoring-ready outputs to support real-time fraud analysis and future system integration.
- Explored responsible GenAI use for feature exploration, anomaly explanation and technical documentation support.

Real-Time Cyber Threat Detection System
- Built a Python-based cyber threat detection system using VirusTotal API for automated URL/file intelligence checks.
- Achieved 98% detection accuracy on evaluated threat samples by combining API signals with classification logic.
- Applied MITRE ATT&CK framework concepts, improving threat pattern classification accuracy by 15%.
- Reduced incident response time by 30% through automated analysis, structured outputs and repeatable checks.
- Implemented API request/response handling, JSON parsing and result formatting for threat reporting.
- Designed modular components to support future updates, additional indicators and workflow scaling.
- Documented threat categories, analysis logic and outputs for technical and non-technical review.

Stock Market Time-Series Forecasting
- Developed LSTM forecasting models for stock market time-series data using historical and real-time datasets.
- Improved prediction accuracy by 10% through preprocessing, experimentation and model tuning.
- Cleaned and transformed financial data into sequential model inputs for trend forecasting.
- Analysed stock movement, patterns and performance indicators using Python-based exploratory analysis.
- Created line charts and trend visualisations to communicate forecasting behaviour and market patterns.
- Built repeatable preprocessing and validation workflows to support consistent time-series modelling.
- Evaluated forecasting outputs by comparing predicted values against actual time-series behaviour.

ATM Fraud Prevention Using Fingerprint Sensor and Blockchain
- Designed a secure ATM fraud prevention concept using fingerprint sensor-based biometric authentication.
- Used blockchain principles to improve transaction transparency, traceability and tamper resistance.
- Created authentication flow logic to verify users before allowing transaction access.
- Applied database concepts to manage user identity, transaction records and validation information.
- Focused on reducing unauthorised access risk through biometric verification and secure transaction logging.
- Prepared system design documentation covering user flow, verification logic and security benefits.
- Built the project as an undergraduate final-year thesis demonstrating secure banking system design.

IPL Statistics Data Analytics Using Kaggle Dataset
- Collected and analysed a large IPL cricket dataset from Kaggle to study players, teams and match outcomes.
- Cleaned missing values, duplicate records, inconsistent formats and irrelevant fields before analysis.

- Performed exploratory data analysis on batting, bowling, match results and season-wise performance trends.
- Created charts to compare players, teams, scores, wickets and performance indicators across seasons.
- Used Pandas and NumPy to group, aggregate, filter and transform cricket statistics into useful insights.
- Presented sports analytics findings through visual summaries suitable for non-technical audiences.
- Documented data cleaning steps and analysis logic to make the workflow repeatable and easy to review.

Greenhouse Effect Data Visualisation

- Analysed environmental datasets related to greenhouse gases, climate patterns, emissions and temperature trends.
- Cleaned and structured raw environmental data to prepare it for visual analysis and interpretation.
- Created charts showing relationships between emissions, temperature changes and environmental impact.
- Used trend analysis to identify long-term climate-related changes and visual patterns in the data.
- Applied Matplotlib and Seaborn to build clear line charts, bar charts and comparative visualisations.
- Converted complex environmental data into understandable visuals for academic and non-technical audiences.
- Connected the project to Green Computing concepts by showing how data can support sustainability awareness.

Underwater Fish Image Classification

- Used an underwater fish image dataset to classify fish categories and analyse image-based patterns.
- Performed image preprocessing including resizing, cleaning and preparation for model training workflows.
- Analysed class frequency distribution to understand dataset balance and category representation.
- Evaluated model performance using precision, recall and F1-score to measure classification reliability.
- Used OpenCV and TensorFlow/Keras concepts to support image handling and deep learning experimentation.
- Applied Scikit-learn metrics to interpret results and compare model performance across classes.
- Documented the workflow from dataset preparation to evaluation to demonstrate practical computer vision knowledge.

Kidney Disease Prediction Web Application

- Built a deep learning-based web application using a large kidney disease dataset for risk prediction.
- Cleaned healthcare data by handling missing values, normalising features and preparing train/test datasets.
- Developed model logic to predict disease risk from structured patient-related input features.
- Integrated the trained model into a basic website interface for user input and prediction output.
- Used TensorFlow/Keras and Scikit-learn for model development, evaluation and performance checking.
- Designed the web layer using Flask/Django basics with HTML and CSS for a simple prediction interface.
- Focused on healthcare analytics by showing how AI can support early risk identification and decision support.

Machine Learning Regression Models on Structured Datasets

- Implemented multiple regression algorithms on structured datasets to predict continuous numerical outcomes.
- Applied Linear Regression, Multiple Regression, Polynomial Regression, Decision Tree Regression and Random Forest Regression.
- Cleaned and prepared datasets using missing value handling, feature selection, encoding and scaling techniques.
- Evaluated models using MAE, MSE, RMSE and R-squared to compare prediction quality.
- Used train-test splitting and model comparison to identify the strongest algorithm for each dataset.
- Created visualisations to explain actual versus predicted values and model behaviour.
- Documented results, limitations and performance differences to strengthen understanding of predictive analytics.

SKILLS

Programming & Data Tools: Advanced Excel, Git/GitHub, Java, Jupyter Notebook, Power BI, Python, R, RStudio, SQL, Tableau

Data Analysis & BI: dashboarding, Data cleaning, data validation, data visualisation, EDA, insight communication, KPI reporting, MI reporting

Machine Learning & AI: classification, CNN-LSTM, feature engineering, LSTM, model evaluation, NLP/LLM awareness, predictive modelling, Regression

Libraries & Frameworks: Flask/Django basics, Matplotlib, NumPy, OpenCV, Pandas, Scikit-learn, Seaborn, TensorFlow/Keras

Data Engineering & Cloud: API integration, AWS S3/EMR basics, data pipelines, ETL concepts, PySpark/Spark basics, scalable preprocessing

Prompt Engineering: documentation support, LLM-assisted analysis, output validation, prompt refinement, responsible AI usage, Structured prompting

Business & Professional: communication, leadership, process analysis, Requirements gathering, stakeholder collaboration, time management, workflow documentation